ESTIM: Efficient and Scalable Tensorial Incomplete Multi-view Semi-supervised Classification

Tingjin Luo
College of Science, National University of Defense Technology
Changsha, China
tingjinluo@hotmail.com

Xiangyao Li
College of Science, National University of Defense Technology
Changsha, China
xiangyaoli2025@hotmail.com

Zhangqi Jiang
College of Science, National University of Defense Technology
Changsha, China
sxdxjzq@gmail.com

Shuanghui Zhang
School of Electronic Science, National University of Defense Technology
Changsha, China
shzhang3@126.com

Dewen Hu
College of Intelligence Science and Technology, National University of Defense Technology
Changsha, China
dwhu@nudt.edu.cn

Abstract

Due to hardware limitations, the dual missing problem with absent views and scarce labels often occurs in multi-view semi-supervised learning (MvSSL), challenging their real-world applications. To solve it, traditional tensor-based methods introduced a third-order tensor to recover the local structures of missing data and existing MvSSL methods have adopted label propagation to enhance the label information of unlabeled data. However, the computational complexity of these methods is cubically dependent on the scale of instances, which severely hampers the scalability to large-scale data. To address these issues, we propose an Efficient and Scalable Tensorial Incomplete Multi-view semi-supervised classification named ESTIM. Specifically, we first replace sample-level graphs with compact bipartite graphs, and then design tensorial graph imputation to explore high-order cross-view information and recover the missing values. Besides, a consensus bipartite graph fusion scheme is designed to align bipartite graphs across views and learn a robust consensus graph against the distorted local structure in label propagation. Moreover, to solve the formulated objective, we derive an efficient optimization algorithm with theoretically proved convergence, whose time complexity is sublinear to the sample sizes. Finally, experimental results across diverse datasets demonstrate the superiority of our ESTIM.

CCS Concepts

• **Computing methodologies** → **Semi-supervised learning settings**; **Classification and regression trees**; **Spectral methods**.

Keywords

Incomplete Multi-view Learning, Graph-based Semi-supervised Classification, Weakly Supervised Learning



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1 Introduction

Multi-view data [11, 19, 42] integrates complementary information from heterogeneous sources, such as visual, textual, and sensor modalities, which can be used to improve the performance of downstream tasks, *i.e.*, medical imaging [17] and autonomous systems [4, 5]. In practice, the inherent heterogeneity of multi-view data complicates the labeling process and increases the cost of label acquisition, resulting in pervasive label scarcity. For example, medical diagnoses require that multiple physicians consider the information for completion [32, 33]. Under the manifold smoothness assumption [46] that samples sharing the same label are likely to lie on the same manifold, Graph-based Multi-view Semi-Supervised Learning (GMvSSL) can utilize large number of unlabeled data and has attracted extensive research attention and been widely applied in various applications. For example, Cai et al. [3] construct a unified graph to describe sample relationships and enhancing the label information across graphs. Nie et al. [25, 35] propose a multi-view learning method with adaptive neighbors, which simultaneously learns an optimal similarity graph. Therefore, these GMvSSL methods have been widely used in many applications.

However, during the data collection process, certain samples may be missing from specific views, resulting in the incompleteness of multiview data sets [36]. This incompleteness makes the smooth manifold assumption difficult to hold in practice. For instance, in multi-sensor surveillance systems, multiple cameras record the same scene simultaneously from various angles. Yet, due to natural or human-induced factors, some sensors may temporarily malfunction, leading to multi-view samples with certain missing views. Web image retrieval presents another such scenario. Thus, efficiently classifying partial multi-view data in dual missing issues, *i.e.* missing views and scarce labels has become a practical and

significant challenge. Some preliminary investigations [34, 38, 41] focus on leveraging the consistencies between multiple modalities, and learn a common similarity graph for label propagation. For example, Trivedi et al. [38] developed a partial modal approach, employing a single modal kernel matrix as the similarity matrix and using Laplacian regularization to impute the missing modal kernel. The limitation of these methods lies in neglecting the representation distortion caused by missing views, which easily violated the underlying smoothness assumptions [24].

To recover the manifold structure of the original data, Tensorial Multi-view Learning [10, 16, 45, 47] approaches are proposed to exploit the high-order correlation among multiple views. In general, these algorithms integrate view-specific representation matrices into a third-order tensor and impose the low-rank tensor constraint to discover consistent structures of data and help to recover its missing values. For instance, Zhou et al. [10, 47] introduced a tensor average rank or Tensor Nuclear Norm (TNN) based on t-SVD to approximate tensor rank and explore the high-order interview correlations. Although they have achieved performance improvements, these methods suffer from computational and spatial complexities that grow quadratically or cubically with the number of samples, resulting in difficulties when applying to large-scale datasets. Besides, the dual missing problem with absent views and scarce labels in GMvSSL often induce discontinuous local structures in each view and the unaligned problem of multi-view data, which would distort graph fusion and further exacerbates the performance degradation.

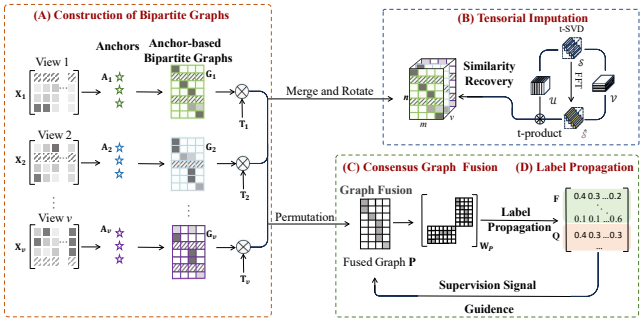


Figure 1: Illustration the main framework of our ESTIM.

To address the above issues, we propose an Efficient and Scalable Tensorial Incomplete Multi-view semi-supervised classification method, named ESTIM, as shown in Figure 1. Specifically, for the prevalent yet underexplored challenge of missing views, we employ a strategy based on incomplete graph matrices constructed for each view. This enables sample classification within each view and concurrently measures the discrepancy between view-specific outcomes and a common class indicator, thereby learning a consolidated probabilistic label matrix across all views. Besides, to overcome the limitations of existing work, which include insufficient exploitation of local structure and high computational costs, we introduce an anchor-based strategy for efficient similarity computation and an efficient optimization algorithm that together reduce computational overhead. To verify the effectiveness of ESTIM, we conduct empirical experiments on multiple public datasets with different view missing and label annotation ratios. Comprehensive experimental results across diverse datasets demonstrate ESTIM outperforms the compared state-of-the-art competitors. Our contributions are summarized as follows:

- We propose a unified model ESTIM that integrates anchor graph learning with tensorial imputation to enhance robustness under severe view and label absence.
- We develop a tensorial imputation approach with bipartite graphs to recover the smooth local structure of missing views and capture high-order cross-view consistency for imputation and consensus graph fusion.
- We design the anchor-based graph fusion operator for view-specific bipartite similarity graphs with anchors ($m \ll n$), which significantly reduce the computational cost of tensorial imputation, enabling scalability to large-scale data.
- An efficient optimization algorithm is derived to solve its objective with theoretical convergence. Extensive experimental results demonstrate its superiority and effectiveness.

2 Preliminary and Related Work

2.1 Notations

Assume $\mathbf{X}=[\mathbf{x}_1,\ldots,\mathbf{x}_n]^\top\in\mathbb{R}^{n\times d}$ is a single view data matrix of d dimensional with n samples, where the first $\ell(\ll n)$ samples are labeled in c classes as $\{y_i\}_{i=1}^{\ell}$. For the sake of calculation, a one-hot label matrix $\mathbf{Y}\in\{0,1\}^{n\times c}$ is used to describe the situation of labels, *i.e.*, $Y_{ij}=1$ if $y_i=j(i\leq\ell)$ and $Y_{ij}=0$ otherwise. Considering the incomplete multi-view data, let $\mathbf{O}^{(v)}\in\{0,1\}^{n\times n}$ be the diagonal indicator matrix of the "existing/missing" state in the v-th view, where $o_{ii}^{(v)}=0$ denotes i-th instance missing its v-th view features and $o_{ii}^{(v)}=1$ otherwise. Additional notations are summarized in Appendix A.

2.2 Third-order Tensor for Completion

To address the view-missing problem, multi-view learning approaches based on tensors have emerged [10, 27]. These methods exploit high-order correlations among views to recover the smooth local structure of the common manifold. Specifically, they construct a third-order tensor by stacking view-specific representation matrices and impose low-rank constraints on this tensor. Denote $\mathcal{X}$ as the original multi-view data tensor, these low-rank constraints through the following optimization procedure:

$$\begin{cases}\min\limits_{\mathcal{G},\mathcal{E}} \mathcal{L}(\mathcal{G})+\lambda\mathcal{R}(\mathcal{E}) \\ \text{s.t. } \mathcal{X}=\mathcal{A}*\mathcal{G}+\mathcal{E},\end{cases} \tag{1}$$

where $\lambda>0$ is the regularization parameter, $\mathcal{L}(\mathcal{G})$ represents the rank of the tensor $\mathcal{G}$ and $\mathcal{R}(\mathcal{E})$ enforces sparsity regularization on the noise component. The noise tensor $\mathcal{E}$ shares the same dimensions. Each frontal slice $\mathbf{A}^{(k)}$ of the core tensor $\mathcal{A}$ constitutes an orthonormal basis matrix for the k-th feature subspace.

As stated in [7], the minimization of the tensor rank in Eq. (1) has been proven to be NP-hard. Therefore, existing methods focus primarily on leveraging surrogate functions to approximate the tensor rank. For example, Wu et al. [10, 39] construct a tensor from multi-view transition probability matrices, applying Singular Value Decomposition (t-SVD) based tensor nuclear norm (TNN) to capture high-order information for clustering task. In this paper, we integrates a third-order bipartite graph tensor and TNN to exploit

high-order cross-view correlations and recover the local structure by imputing its missing values for GMvSSL.

2.3 Multi-view Semi-supervised Learning

To overcome the high labeling expenses inherent in multi-view data, several research [14, 21] have increasingly focused on multi-view semi-supervised learning. By synergistically leveraging multi-view complementary information and semi-supervised learning mechanisms, significantly enhancing the classification performance.

Among emerging methodologies, GMvSSL forms a prominent approach. Lu et al. [22] propose Generative Essential Graph Convolutional Network (GEGCN) that preserves view-specific topological structures while enforcing consistency through spectral convolution operator. concurrently, Huang et al. [9] introduce a geometric breakthrough with their Geometric Localized Graph Convolutional Network (GL-GCN), which explicitly models the manifold structure through diffusion mapping. Beyond these graph-based methods, matrix factorization techniques offer alternative representation learning strategies focusing on latent space decomposition. For example, SMVCF [14] integrates semi-supervised label propagation with manifold regularization to learn discriminative shared concept representations. Despite their success on complete-view benchmarks, the above methods encounter practical limitations when handling missing views in real-world multi-view data.

To address it, most recently, Zhuge et al. [49] propose a framework, termed AMSC, for incomplete multi-view semi-supervised learning constructs partial similarity graphs from available views, propagates labels to generate view-specific matrices, and fuses them into a consensus label matrix. Despite adaptive fusion via p-th root weighting, AMSC fails to address structural distortions caused by missing views in graph manifolds.

3 Methodology

3.1 Proposed Formulation

To handle missing views problem, current approaches [10, 45, 47] stack view-specific similarity graphs into a third-order tensor to reconstruct missing samples' relationships with multi-view consistent correlations. However, tensor decomposition exhibits $O(n^3)$ time complexity, rendering it infeasible for large-scale datasets. To overcome $O(n^3)$ complexity barrier, we propose ESTIM which adopts an anchor acceleration strategy for dimensionality reduction to deliver accelerated classification with $O(n\log(n))$ complexity. The structure of our ESTIM includes four parts: (1) construction of bipartite graphs, (2) tensorial imputation, (3) consensus graph fusion and (4) label enhancement with propagation.

A. Construction of Bipartite Graphs

Let$\{\mathbf{X}_v \in \mathbb{R}^{d_v\times n}\}_{v=1}^V$ denote a multi-view dataset with n samples, where V is the number of views, d_v is the feature dimension of the v-th view. Unlike previous works [20, 29] that fuse sample-level similarity matrices, we introduce anchor-based graphs $\mathbf{G}_v$ to accelerate tensor learning. By extracting a compact anchor set ($m \ll n$) from samples and building view-specific affinity graphs $\mathbf{G}_v \in \mathbb{R}^{n\times m}$ ($g_{ij}^{(v)} \geq 0$) , we efficiently model the underlying manifold structures. The bipartite graphs $\mathbf{G}_v$ are obtained by solving:

$$\begin{cases} \min\limits_{\mathbf{G}_v} \sum\limits_{i=1}^{n}\sum\limits_{j=1}^{m}\left(\left\|\mathbf{x}_i^{(v)}-\mathbf{a}_j^{(v)}\right\|_2^2 g_{ij}^{(v)}+\gamma\left(g_{ij}^{(v)}\right)^2\right), \\ \text{s.t. } \mathbf{G}_v\mathbf{1}_m=\mathbf{1}_n, \mathbf{G}_v\geq 0, \end{cases} \tag{2}$$

where $\mathbf{x}_i^{(v)}$ and $\mathbf{a}_j^{(v)}$ represent the i−th sample and j-th anchor, $g_{ij}^{(v)}$ is the (i,j)-th element of $\mathbf{G}_v$. By using the anchor-graph strategy, the $n\times n$ sample-level similarity graph $\mathbf{W}_v$ in each view can be reduced to the $n\times m$ bipartite graph $\mathbf{G}_v$, which can reduce its computational complexity of tensor imputation. Nevertheless, due to the randomness of the missing view, we can observe that independently constructing the bipartite graph $\mathbf{G}_v$ via Eq. (2) in each view easily causes the selected anchor sets generated from different views to mismatch, as there are no guarantees of correspondences between them, that is the traditional Anchor Unaligned Problem (AUP) reported in [8].

B. Consensus Graph Fusion

To address AUP problem, we introduce the permutation matrices $\{\mathbf{T}_v \in \mathbb{R}^{m\times m}\}_{v=1}^V$ to align the anchors between different views and learn a consensus fused graph $\mathbf{P}\in\mathbb{R}^{n\times m}$ from $\{\mathbf{G}_v\}_{v=1}^V$. Different views contribute to the different importance. In addition, to adaptively explore each view-specific structure information, we assign a learnable weight α_v to each view and derive an adaptive consensus graph fusion (CGF) operator, which is defined as follows:

$$\begin{cases} \min\limits_{\mathbf{P},\alpha}\ \mathtt{CGF}(\mathbf{P},\{\mathbf{G}_v\}_{v=1}^V) \triangleq -\sum\limits_{v=1}^{V}\alpha_v \mathrm{Tr}\left(\mathbf{P}^\top(\mathbf{G}_v\mathbf{T}_v)\right)+\beta\|\mathbf{P}\|_F^2 \\ \text{s.t. } \mathbf{T}_v^\top\mathbf{T}_v=\mathbf{I}_m\ (\forall v), \alpha^\top\alpha=1,\ \mathbf{P}\in\Delta_m^n. \end{cases} \tag{3}$$

where $\mathbf{I}_m\in\mathbb{R}^{m\times m}$ is an identity matrix, $\alpha=[\alpha_1,\ldots,\alpha_V]^\top$, and Δ_m^n $=\{\zeta\in\mathbb{R}^{n\times m}|\zeta\mathbf{1}_m=\mathbf{1}_n,\zeta\geq 0\}$. Noting $\mathbf{G}_v$ in Eq. (2) merely encodes sample-anchor similarities, which leads to the consensus fused graph $\mathbf{P}$ requires additional processing before label propagation.

C. Tensorial Imputation

Due to the missing view, the obtained consensus fused graph $\mathbf{P}$ in Eq. (3) is incomplete and makes AUP problem even more severe. Let ψ_v and ϕ_v index observed and missing samples in view v, the bipartite graph $\mathbf{G}_v$ can be decomposed into $\mathbf{G}_{\phi_v}$ and $\mathbf{G}_{\psi_v}$, where $\mathbf{G}_{\phi_v}$ is built by Eq. (2) for existing samples, and $\mathbf{G}_{\psi_v}$ is initialized as an equal probability matrix. To tackle this issue, we develop a tensorial imputation approach to recover the smooth local structure of missing views and capture high-order cross-view consistency for reconstructing similarity relationships of missing samples. To capture high-order inter-view correlations, we first stack the bipartite graphs $\{\mathbf{G}_v\}_{v=1}^V$ into a tensor $\mathcal{G}=\Phi(\mathbf{G}_1,\ldots,\mathbf{G}_V)\in\mathbb{R}^{m\times V\times n}$. The similarity relationships between missing samples and anchors in each view are imputed via the cross-view consistency information. Therefore, to recover the incomplete graph elements of $\mathcal{G}$, we imposed tensorial imputation with the TNN regularizer into the CGF operator for missing-view scenarios. Thereby, the problem (3) can be reformulated as:

$$\begin{cases} \min\limits_{\mathbf{P},\alpha}\ \mathtt{CGF}(\mathbf{P},\{\mathbf{G}_v\}_{v=1}^V)+\|\mathcal{G}\|_{\circledast} \\ \text{s.t. } \mathbf{T}_v^\top\mathbf{T}_v=\mathbf{I}_m\ (\forall v), \alpha^\top\alpha=1,\ \mathbf{P}\in\Delta_m^n. \end{cases} \tag{4}$$

where $\|\mathcal{G}\|_{\circledast}$ are defined in Appendix A.

D. Label Enhancement with Propagation

Motivated by [43], to better leverage the local structural information across neighbors, we construct the sample-level graph with $\mathbf{P}$ as $\mathbf{W}_P=\begin{bmatrix} & \mathbf{P} \\ \mathbf{P}^\top & \end{bmatrix} \in \mathbb{R}^{(n+m)\times(n+m)}$. After obtaining the sample-level graph $\mathbf{W}_p$, similar with [23, 28], the objective function of label enhancement for the unlabeled instances can be formulated as:

$$\mathcal{L}(\mathbf{F}, \mathbf{L}_W) = \mathrm{Tr}(\mathbf{F}^\top \mathbf{L}_W \mathbf{F}) + \mathrm{Tr}((\mathbf{F}-\mathbf{Y})^\top \mathbf{D}_W (\mathbf{F}-\mathbf{Y})), \tag{5}$$

where, $\mathbf{Y} \in \{0,1\}^{n\times c}$ encodes the one-hot label matrix for n samples across c classes, $\mathbf{F} \in \mathbb{R}^{n\times c}$ is a soft label matrix, $\mathbf{D}_W$ is a diagonal matrix, and $\mathbf{L}_W = \mathbf{D}_W - \mathbf{W}$ is a Laplacian matrix associated with $\mathbf{W}$. To maintain dimensional consistency when substituting $\mathbf{W}_P$ into Eq. (5), we incorporate a soft label matrix of fused anchors $\mathbf{Q} \in \mathbb{R}^{m\times c}$. Thus, the loss function with the normalized Laplacian matrix $\tilde{\mathbf{L}}_{W_P}=\mathbf{I}_{n+m}-\mathbf{D}_{W_P}^{-\frac{1}{2}}\mathbf{W}_P\mathbf{D}_{W_P}^{-\frac{1}{2}}=\mathbf{I}_{n+m}-\mathbf{W}_P$ can be rewritten as:

$$\begin{aligned}\mathcal{L}(\hat{\mathbf{F}}, \mathbf{L}_{W_P}) &= \mathrm{Tr}(\hat{\mathbf{F}}^\top \mathbf{L}_{W_P} \hat{\mathbf{F}}) + \mathrm{Tr}\left((\hat{\mathbf{F}}-\hat{\mathbf{Y}})^\top \mathbf{D}_{W_P}(\hat{\mathbf{F}}-\hat{\mathbf{Y}})\right) \\ &= \mathrm{Tr}\left((\mathbf{I}_{n+m}+\mathbf{D}_{W_P})\hat{\mathbf{F}}^\top\hat{\mathbf{F}}\right) - \mathrm{Tr}\left(\hat{\mathbf{F}}^\top\mathbf{W}_P\hat{\mathbf{F}}\right) - 2\mathrm{Tr}\left(\mathbf{D}_{W_P}\hat{\mathbf{Y}}\hat{\mathbf{F}}^\top\right),\end{aligned} \tag{6}$$

where $\hat{\mathbf{F}} = [\mathbf{F}; \mathbf{Q}] \in \mathbb{R}^{(n+m)\times c}$ and $\hat{\mathbf{Y}} = [\mathbf{Y}; \mathbf{0}]$ denote vertically concatenated matrices, $\mathbf{D}_{W_P}$ is the degree matrix, and $\mathbf{L}_{W_P} = \mathbf{D}_{W_P} - \mathbf{W}_P$ is the Laplacian matrix for $\mathbf{W}_P$. Finally, incorporating Eq. (4) into Eq. (6), the final objective of ESTIM can be expressed as:

$$\begin{cases} \min\limits_{\{\mathbf{G}_{\psi_v},\mathbf{T}_v\}_{v=1}^V,\hat{\mathbf{F}},\,\alpha,\,\mathbf{P}} \mathcal{L}(\hat{\mathbf{F}}, \mathbf{L}_{W_P}) + \lambda\mathtt{CGF}(\mathbf{P}, \{\mathbf{G}_v\}_{v=1}^V) + \rho\|\mathcal{G}\|_\circledast \\ \text{s.t. } \mathbf{P}, \mathbf{G}_v \in \Delta_m^n, \mathbf{T}_v^\top\mathbf{T}_v = \mathbf{I}_m \ (\forall v \in [V]), \alpha^\top\alpha = 1, \mathcal{G} = \Phi(\mathbf{G}_1, \ldots, \mathbf{G}_V). \end{cases} \tag{7}$$

where λ and ρ are regularization parameters.

3.2 Optimization

To solve the variables in Eq. (7), inspired by ADMM [2], we introduce an auxiliary tensor variable $\mathcal{H}$ to relax $\mathcal{G}$, so the objective function of Eq. (7) can be rewritten as:

$$\begin{aligned}&\mathcal{L}\left(\{\mathbf{G}_{\psi_v},\ \mathbf{T}_v\}_v^V, \hat{\mathbf{F}},\ \alpha,\ \mathbf{P},\ \mathcal{H},\ \mathcal{U}\right) \\ &= \mathrm{Tr}\left((\mathbf{I}_{n+m}+\mathbf{D}_{W_P})\hat{\mathbf{F}}^\top\hat{\mathbf{F}}\right) - \mathrm{Tr}\left(\hat{\mathbf{F}}^\top\mathbf{W}_P\hat{\mathbf{F}}\right) \\ &- 2\mathrm{Tr}\left(\mathbf{D}_{W_P}\hat{\mathbf{Y}}\hat{\mathbf{F}}^\top\right) - \lambda\sum\nolimits_v \alpha_v \mathrm{Tr}\left(\mathbf{P}^\top(\mathbf{G}_v\mathbf{T}_v)\right) \\ &+ \beta_\lambda\|\mathbf{P}\|_F^2 + \rho\|\mathcal{H}\|_\circledast + \langle\mathcal{U}, \mathcal{G}-\mathcal{H}\rangle + \frac{\eta}{2}\|\mathcal{G}-\mathcal{H}\|_F^2,\end{aligned} \tag{8}$$

where $\beta_\lambda=\lambda\cdot\beta$, $\mathcal{U}$ is Lagrange multiplier, and $\eta>0$ is a nonnegative penalty parameter. Then, the optimization problem (8) can be decomposed into five subproblems, each of which optimizes its respective variables independently while keeping others fixed.

• $\mathbf{G}_{\psi_v}$**-Subproblem:** By fixing the remaining variables, we arrive at the following problem formulation for $\mathbf{G}_v$:

$$\begin{cases} \min\limits_{\mathbf{G}_{\psi_v}} \frac{\eta}{2}\|\mathbf{G}_v - \mathbf{H}_v\|_F^2 + \langle\mathbf{U}_v, \mathbf{G}_v - \mathbf{H}_v\rangle - \lambda\alpha_v \mathrm{Tr}\left(\mathbf{P}^\top(\mathbf{G}_v\mathbf{T}_v)\right), \\ \text{s.t. } \mathbf{G}_{\psi_v} \in \Delta_m^{|\psi_v|}, \end{cases} \tag{9}$$

Eq. (9) can be rewritten as the following element-wise form:

$$\begin{aligned}\min_{\mathbf{G}_{\psi_v}\in\Delta_m^{|\psi_v|}} &\sum\nolimits_{i\in\psi_v}\sum\nolimits_{j=1}^m \frac{\eta}{2}(G_{ij}^{(v)} - H_{ij}^{(v)})^2 \\ &+ U_{ij}^{(v)}(G_{ij}^{(v)} - H_{ij}^{(v)}) - \lambda\alpha_v(\mathbf{P}\mathbf{T}_v^\top)_{ij}G_{ij}^{(v)},\end{aligned} \tag{10}$$

According to [30], we introduce a Lagrange multiplier ζ to merge equality constraints into the objective function, forming an unconstrained Lagrangian function $L(\mathbf{G}_{i\cdot}^{(v)}, \zeta, \boldsymbol{\epsilon}_i)$. This formulation enables us to obtain closed-form solutions.

$$G_{ij}^{(v)\star} = \left(H_{ij}^{(v)} - \frac{2}{\eta}G_{ij}^{(v)} + \zeta\right)_+. \tag{11}$$

Each $\mathbf{G}_{i\cdot}^{(v)}$, $i\in\psi_v$ can then be solved and we can update $\mathbf{G}_{\psi_v}$.

• **P-Subproblem:** By fixing the other variables, we derive a min optimization problem:

$$\begin{cases} \min\limits_{\mathbf{P}} -\lambda\sum\limits_v \alpha_v \mathrm{Tr}\left(\mathbf{P}^\top\mathbf{G}_v\mathbf{T}_v\right) + \beta_\lambda\|\mathbf{P}\|_F^2 - \mathrm{Tr}\left(\hat{\mathbf{F}}^\top\mathbf{W}_P\hat{\mathbf{F}}\right), \\ \text{s.t. } \mathbf{P}\in\Delta_m^n, \end{cases} \tag{12}$$

recall that $\hat{\mathbf{F}}=[\mathbf{F};\mathbf{Q}]$, $\mathrm{Tr}(\hat{\mathbf{F}}^\top\mathbf{W}_P\hat{\mathbf{F}}) = \sum_i^n\sum_j^m \left\langle \mathbf{F}_{i,}, \mathbf{Q}_{j,}\right\rangle P_{ij}$. Similar to the subproblem of $\mathbf{G}_{\psi_v}$, we can obtain the optimal $\mathbf{P}^\star$ of the problem by solving the following proximal problem [30]:

$$\begin{cases} \arg\min\limits_{\mathbf{P}_{i\cdot}} \frac{1}{2}\left\|\mathbf{P}_{i\cdot} - \frac{1}{\beta_\lambda}\left(\frac{\lambda}{2}\tilde{\mathbf{G}}_{i\cdot} + \mathbf{K}_{i\cdot}\right)\right\|_F^2, \\ \text{s.t. } \mathbf{P}_{i\cdot}\mathbf{1}_m = 1, \mathbf{P}_{i\cdot} \geq 0, \end{cases} \tag{13}$$

where matrix $\tilde{\mathbf{G}}=\sum_v \alpha_v\mathbf{G}_v\mathbf{T}_v$ and matrix $\mathbf{H}\in\mathbb{R}^{n\times m}$ measures the distance between the soft labels of $\mathbf{F}$ and $\mathbf{Q}$ with its element $K_{ij}=\left\langle \mathbf{F}_{i,}, \mathbf{Q}_{j,}\right\rangle$. Since the feasible region of Eq. (13) is a closed convex set x, the Hilbert projection theorem guarantees the uniqueness of the optimal solution $\mathbf{P}^\star$. Denote $\mathbf{C}_{i\cdot} = \frac{\lambda\tilde{\mathbf{G}}_{i\cdot}+\mathbf{K}_{i\cdot}}{2\beta\lambda}$, the Lagrangian function of problem (14) can be written as :

$$L(\mathbf{P}_{i\cdot}, \eta, \boldsymbol{\beta}_i) = \frac{1}{2}\|\mathbf{P}_{i\cdot} - \mathbf{C}_{i\cdot}\|_2^2 - \eta(\mathbf{P}_{i\cdot}\mathbf{I}_m - 1) - \mathbf{P}_{i\cdot}\boldsymbol{\beta}_i, \tag{14}$$

where η and $\boldsymbol{\beta}_i \geq 0$ are the Lagrange multipliers, according to the KKT condition. Then, we have:

$$P_{ij} = (C_{ij} + \eta)_+. \tag{15}$$

• $\hat{\mathbf{F}}$**-Subproblem:** Fixing the other variables leads to the following problem for $\hat{\mathbf{F}}$,

$$\min_{\hat{\mathbf{F}}} \mathrm{Tr}\left((\mathbf{I}_{n+m}+\mathbf{D}_{W_P})\hat{\mathbf{F}}^\top\hat{\mathbf{F}}\right) - \mathrm{Tr}\left(\hat{\mathbf{F}}^\top\mathbf{W}_P\hat{\mathbf{F}}\right) - 2\mathrm{Tr}\left(\mathbf{D}_{W_P}\hat{\mathbf{Y}}\hat{\mathbf{F}}^\top\right), \tag{16}$$

By setting the derivative of problem (16) to zero, $\hat{\mathbf{F}}$ can be updated as $\hat{\mathbf{F}}^* = (\mathbf{I}_{n+m}+\mathbf{D}_{W_P}-\mathbf{W}_P)^{-1}\mathbf{D}_{W_P}\hat{\mathbf{Y}}$. Recall that $\mathbf{W}_P=\begin{bmatrix} & \mathbf{P} \\ \mathbf{P}^\top & \end{bmatrix}$, since $\mathbf{W}_P$ and $\mathbf{D}_{W_P}$ are both block matrices, we can adopt the block matrix inversion to solve the first term:

$$\begin{aligned}(\mathbf{I}_{n+m}+\hat{\mathbf{D}}-\mathbf{W}_P)^{-1} &= \begin{bmatrix} \mathbf{I}_n+\mathbf{D}_n & -\mathbf{P} \\ -\mathbf{P}^\top & \mathbf{I}_m+\mathbf{D}_m, \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \mathbf{C}_1^{-1} & -\mathbf{M}_{11}^{-1}\mathbf{M}_{12}\mathbf{C}_2^{-1} \\ -\mathbf{C}_2^{-1}\mathbf{M}_{21}\mathbf{M}_{11}^{-1} & \mathbf{C}_2^{-1} \end{bmatrix},\end{aligned} \tag{17}$$

where $\mathbf{M}_{11}=\mathbf{I}_n+\mathbf{D}_n$, $\mathbf{C}_1=\mathbf{M}_{11}-\mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21}$, $\mathbf{M}_{12}=-\mathbf{P}$, $\mathbf{M}_{22}=\mathbf{I}_m+\mathbf{D}_m$, $\mathbf{C}_2=\mathbf{M}_{22}-\mathbf{M}_{21}\mathbf{M}_{11}^{-1}\mathbf{M}_{12}$, and $\mathbf{M}_{21}=\mathbf{P}^\top$. Since $\mathbf{C}_1\in\mathbb{R}^{n\times n}$ needs time complexity $O(n^3)$ to solve $\mathbf{C}_1^{-1}$, we utilize the Woodbury matrix identity to accelerate the inversion by $\mathbf{C}_1^{-1}=\mathbf{M}_{11}^{-1}+\mathbf{M}_{11}^{-1}\mathbf{M}_{12}(\mathbf{M}_{22}-\mathbf{M}_{21}\mathbf{M}_{11}^{-1}\mathbf{M}_{12})^{-1}\mathbf{M}_{21}\mathbf{M}_{11}^{-1}$, which reduces $O(n^3)$ to $O(nm^2)$. Note that $\mathbf{M}_{11}\in\mathbb{R}^{n\times n}$, but it is a diagonal matrix and its inversion can be easily obtained with $O(n)$. Then, leveraging the block matrix inversion in Eq. (17), the update formula for $\hat{\mathbf{F}}^\star=[\mathbf{F}^\star;\mathbf{Q}^\star]$ can be divided into two parts $\mathbf{F}$ and $\mathbf{Q}$, respectively:

$$\begin{cases}\mathbf{F}^\star=\mathbf{M}_{11}^{-1}\mathbf{M}_{12}(\mathbf{M}_{22}-\mathbf{M}_{21}\mathbf{M}_{11}^{-1}\mathbf{M}_{12})^{-1}\mathbf{M}_{21}\mathbf{M}_{11}^{-1}\mathbf{B}_n\mathbf{Y}+\mathbf{M}_{11}^{-1}\mathbf{B}_n\mathbf{Y},\\ \mathbf{Q}^\star=-(\mathbf{M}_{22}-\mathbf{M}_{21}\mathbf{M}_{11}^{-1}\mathbf{M}_{12})^{-1}\mathbf{M}_{21}\mathbf{M}_{11}^{-1}\mathbf{B}_n\mathbf{Y}.\end{cases} \tag{18}$$

• $\mathcal{H}$**-Subproblem:** When other variables are fixed, the subproblem for $\mathcal{H}$ is formulated as:

$$\min_{\mathcal{H}}\ \frac{\rho}{\eta}\|\mathcal{H}\|_{\circledast}+\frac{1}{2}\left\|\mathcal{H}-(\mathcal{G}+\frac{\mathcal{J}}{\eta})\right\|_F^2. \tag{19}$$

The subproblem in Eq. (19) can be solved by the following theorem.

Theorem 3.1. [40] *Suppose $\mathcal{H},\mathcal{F}\in\mathbb{R}^{n_1\times n_2\times n_3}$ and $\tau>0$, the globally optimal solution to $\min_{\mathcal{H}}\tau\|\mathcal{H}\|_{\circledast}+\frac{1}{2}\|\mathcal{H}-\mathcal{F}\|_F^2$ is given by the tensor tubal-shrinkage operator, i.e., $\mathcal{H}=\mathcal{C}_{n_3\tau}(\mathcal{F})=\mathcal{U}*\mathcal{C}_{n_3\tau}(\mathcal{S})*\mathcal{V}^\top$, where $\mathcal{F}=\mathcal{U}*\mathcal{S}*\mathcal{V}^\top$ is obtained by t-SVD and $\mathcal{C}_{n_3\tau}(\mathcal{S})=\mathcal{S}*\mathcal{J}$, herein, $\mathcal{J}$ is an $n_1\times n_2\times n_3$ f-diagonal tensor whose diagonal element in the Fourier domain is $\mathcal{J}_f^k(i,i)=(1-n_3\tau/\mathcal{S}_f^k(i,i))_+$.*

• $\mathbf{T}_v$**-Subproblem:** By fixing the other variables, $\mathbf{T}_v$ can be independently updated by,

$$\mathbf{T}_v^\star=\arg\max_{\mathbf{T}_v}\ \mathrm{Tr}(\mathbf{T}_v^\top\mathbf{Z}_v^\top\mathbf{P}),\ \text{s.t. }\mathbf{T}_v^\top\mathbf{T}_v=\mathbf{I}_m, \tag{20}$$

Given Singular Value Decomposition (SVD) of the product $\mathbf{G}_v^\top\mathbf{P}=\mathbf{U}_v\Sigma_v\mathbf{V}_v^\top$, the objective function Eq. 20 can be formulated as:

$$\mathrm{Tr}\left(\mathbf{T}_v^\top\mathbf{U}_v\Sigma_v\mathbf{V}_v^\top\right)=\mathrm{Tr}\left(\mathbf{V}_v^\top\mathbf{T}_v^\top\mathbf{U}_v\Sigma_v\right), \tag{21}$$

Define the orthogonal transformation matrix $\mathbf{E}_v=\mathbf{V}_v^\top\mathbf{T}_v^\top\mathbf{U}_v$, which satisfies the orthogonality constraint $\mathbf{E}_v\mathbf{E}_v^\top=\mathbf{I}$, we obtain

$$\mathrm{Tr}\left(\mathbf{V}_v^\top\mathbf{T}_v^\top\mathbf{U}_v\Sigma_v\right)=\mathrm{Tr}\left(\mathbf{E}_v\Sigma_v\right)\le\sum_{i=1}^{m}\sigma_i^{(v)}, \tag{22}$$

where $\sigma_i^{(v)}$ denotes the i-th singular value along the diagonal of Σ_v. This inequality follows from von Neumann's trace theorem. The upper bound is achieved when $\mathbf{E}_v=\mathbf{I}$, corresponding to the optimal transformation:

$$\mathbf{T}_v^\star=\mathbf{U}_v\mathbf{V}_v^\top. \tag{23}$$

• α**-Subproblem:** When other variables are fixed, the subproblem for α_v is formulated as :

$$\begin{cases}\min\limits_{\alpha_v}-\lambda\sum\limits_v\alpha_v\mathrm{Tr}(\mathbf{P}^\top(\mathbf{Z}_v\mathbf{T}_v)),\\ \text{s.t. }\alpha^\top\alpha=1,\alpha_v\ge0.\end{cases} \tag{24}$$

Denote $R_v=\mathrm{Tr}(\mathbf{P}^\top(\mathbf{Z}_v\mathbf{T}_v))$, the Lagrangian function of problem (24) can be formulated as:

$$L(\boldsymbol{\alpha},\mu,\boldsymbol{\beta})=-\lambda\sum_v R_v\alpha_v+\mu(\sum_v\alpha_v^2-1)-\sum_v\beta_v\alpha_v, \tag{25}$$

where β_v and $\mu\ge0$ are KKT multipliers for non-negativity ($\beta_v\alpha_v=0$) and equality constraint ($\alpha^\top\alpha=1$) respectively. According to the KKT conditions, optimal solution is obtained as,

$$\alpha_v^*=\frac{R_v}{\sqrt{\sum_v R_v^2}}. \tag{26}$$

At last, the Lagrange multipliers and penalty parameters are updated as $\mathcal{U}=\mathcal{U}+\eta(\mathcal{G}-\mathcal{H})$ and $\eta=\min(\gamma_\eta\eta,\eta_{\max})$, respectively, where $\gamma_\eta>1$ is used to accelerate convergence. The detailed algorithm of our method is summarized in Algorithm 1.

Algorithm 1 Optimization Algorithm of ESTIM

Input: Bipartite graphs $\{\mathbf{G}_{\phi_v}\in\mathbb{R}^{|\phi_v|\times m}\}_{v=1}^V$, one-hot label matrix $\mathbf{Y}$, the parameters $\hat{D}$, λ, β, and ρ.
Output: Labels of unlabeled samples $\tilde{\mathbf{Y}}_u$.
1. For each $v\in[V]$ Initialize $\mathbf{G}_{\psi_v}=\frac{1}{m}\mathbf{1}_{|\psi_v|}\mathbf{1}_m^\top$, $\mathbf{T}_v=\mathbf{I}$, $\alpha_v=\frac{1}{V}$.
2. Initialize variables $\mathbf{H}=\mathbf{U}=\mathbf{0}$, $\eta=10^{-2}$, $\gamma_\eta=2$, $\eta_{\max}$.
3. Calculate $\mathbf{P}$ and $\mathbf{F}$ by Eq. (11) and Eq. (15), respectively.
While *not converged* **do**
4. For each $i\in\psi_v$, update the i-th row of $\mathbf{G}$ by Eq. (11).
5. Compute fused graph $\mathbf{P}$ by Eq. (15).
6. Update $\mathbf{F}$ and $\mathbf{Q}$ by Eq. (18).
7. Caculate $\mathbf{H}$ by solving problem (19).
8. For each $v\in[V]$, update $\mathbf{T}_v$ by solving Eq. (20).
9. Update α by solving Eq. (26).
10. Compute $\mathbf{U}\leftarrow\mathbf{U}+\eta(\mathbf{G}-\mathbf{H})$.
11. Update penalty parameter $\eta\leftarrow\min(\gamma_\eta\eta,\eta_{\max})$.
End While
12.**Return** $\tilde{\mathbf{Y}}_u\triangleq\{\tilde{y}_i\mid\tilde{y}_i=\arg\max_{j\in[c]}\mathbf{F}_{ij}\}_{i=\ell+1}^n$.

3.3 Convergence and Complexity Analysis

Convergence. The theoretical convergence guarantee of ESTIM is formally established in Theorem 3.2. As stated in Theorem 3.2 generated by Algorithm 1 tends, the sequence to converge to a Karush-Kuhn-Tucker (KKT) stationary point [1]. The detailed proof of Theorem 3.2 is provided in Appendix B.

Theorem 3.2. *Consider the sequences generated by ESTIM, The sequence satisfies the following two key properties: (1) The sequence $\{L1(k)\}_{k=1}^\infty$ is bounded. (2) Any accumulation point of $\{L1(k)\}_{k=1}^\infty$ is a Karush–Kuhn–Tucker (KKT) critical point.*

Time Complexity Analysis. For the proposed ESTIM, The computational complexity of ESTIM primarily arises from the update of variables: $\mathbf{G}_{\psi_v}$, $\mathbf{P}$, α, $\hat{\mathbf{F}}$, $\mathcal{H}$, and $\mathbf{T}_v$. The update of the variables $\{\mathbf{G}_{\psi_v}\}_{v=1}^V$ requires $O(\sum_v^V|\psi_v|(m^2+m))$. In algorithm 1, the computational complexity for each update of $\mathbf{H}$ and $\tilde{\mathbf{G}}$ is $O(nm(c+1))$ and $O(nm(mV+1))$, respectively. Thus, the time complexity of updating $\mathbf{P}$ is $O(nm(mV+c+2))$. For $\hat{\mathbf{F}}$, it requires $O(nm^2)$ for each iteration. For tensor $\mathcal{H}$, each update needs fast Fourier transformation (FFT), inverse FFT, and t-SVD operations, corresponding to a computational complexity of $O(nmV\log(nV)+nmV^2)$. The update of $\{\mathbf{T}_v\}_{v=1}^V$ requires $O(nm^2+2m^3)$ for each iteration. Assume T iterations are required to achieve convergence. For α, it requires $O(nm^2)$ for each iteration. Considering $m,c,V\ll n$, the overall time complexity of ESTIM is $O(T(nmV\log(nV)+nm^2V+nmV^2))$.

Space Complexity Analysis. For our proposed method, the major memory costs are various anchored matrices and tensors. From the optimization in Section 3.2, the space complexity of ESTIM is $O(n(mV + c + \sum_v d_v))$.

4 Experiments

4.1 Experimental Settings

Data sets. We conducted experiments on eight real multi-view datasets, including CUB, UCI-Digit, Handwritten, MNIST4, Scene15, Caltech101, AwA, and Animal. The main summary are present in Table1 and more detailed information can be found below.

- *CUB* [1] The Caltech-UCSD Birds (CUB) benchmark contains 11,788 images representing 200 avian species. Consistent with prior work [15], we utilize the initial 10 categories, employing 1024-dimensional convolutional features extracted from GoogLeNet alongside 300-dimensional semantic embeddings generated by doc2vec.
- *UCI-Digit* [2] This repository comprises 2,000 handwritten digit samples (0-9) with uniform class distribution. Following by [49], we extract three complementary representations: 76D Fourier shape descriptors, 216D profile correlations, and 64D Karhunen-Love coefficients coefficients.
- *Handwritten* [3] Derived from the UCI multiple features dataset, containing 2,000 handwritten digit samples across 10 categories. Each sample is characterized by six handcrafted features: 240D pixel averages, 216D profile correlations, 76D Fourier coefficients, 64D Karhunen-Loève coefficients, 47D Zernike moments, and 6D morphological features.
- *MNIST4* [4] Derived from the MNIST handwritten digit database, this subset contains the first four digit classes with 4,000 samples. Each sample is characterized by three heterogeneous feature representations: 30D Isometric Projection (ISO), 9D Linear Discriminant Analysis (LDA), and 30D Neighborhood Preserving Embedding (NPE).
- *Scene15* [5] A standard scene classification benchmark consisting of 4,485 images across 15 indoor and outdoor categories. Each image is characterized by four complementary features: 1000D bag-of-words (BoW) with SIFT descriptors, 512D GIST scene descriptors, 256D color histograms, and 2000D deep features extracted from a pre-trained ResNet-50 model.
- *Caltech101* [6] The full Caltech101 object recognition database contains 9,146 images from 101 categories plus a background clutter class. We employ the standard six visual representations: 48D Gabor filter responses, 40D wavelet moments, 512D GIST descriptors, 1984D histogram of oriented gradients (HOG), 928D local binary patterns (LBP), and 4096D VGG-16 deep features.
- *AwA* [7] Containing 30,475 images spanning 50 animal classes. We adopt the standardized subset of 10,158 images [44], represented by dual 4096D deep features from DECAF and VGG19 architectures.
- *Animal* [8] This dataset comprises 11,673 samples spanning 50 animal species, with each sample represented by four complementary feature views: 1024D ResNet-50 visual features, 512D LBP/HOG texture descriptors, 300D Word2Vec semantic embeddings, and 256D geometric shape features.

Table 1: Details of the datasets used in our experiments

Feature type	CUB	UCI-Digit	Handwritten	MNIST4
1	VIS(1024)	FOU(64)	Pixel(240)	ISO(30)
2	ATT(300)	FAC(76)	Fourier(76)	LDA(9)
3	-	KAR(216)	Profile(216)	NPE(30)
4	-	-	Zernike(47)	-
5	-	-	KAR(64)	-
6	-	-	Morph(6)	-
# of Samples	600	2,000	2,000	4,000
# of Classes	10	10	10	4
# of numbers	64	256	256	256
Feature type	Scene15	Caltech	AwA	Animal
1	SPE(20)	GAB(48)	GIST(512)	VIS(2689)
2	LBP(59)	WM(40)	LBP(928)	ATT(2000)
3	CH(40)	CENT(254)	-	TEX(2000)
4	-	-	-	GEX(2000)
# of Samples	4,485	9,144	10,158	11673
# of Classes	15	101	50	20
# of numbers	256	512	512	512

Baselines. To validate the effectiveness of ESTIM, we compare it with the following methods. First, we select SLIM [41] and AMSC [49] as two representative state-of-the-art incomplete multi-view semi-supervised learning (IMvSSL) baselines. In addition, we also compare it with several GMvSSL methods, *i.e.*, AMGL [26], MLAN [25], FMSSL [43], FMSEL [20], LPvote [48] and CFMSC [12]. We also incorporate popular regression-based MvSSL methods, *i.e.*, **MVAR** [37] and SMIR [31]. Most baselines, except **SLIM** and **AMSC**, cannot handle the incomplete data. For fair comparison, following [49], we employ deep matrix factorization (DMF) [6] to impute incomplete feature matrices.

Parameter settings. Following [13], we select m anchor points per view using the BKHK or k-means algorithm, where m is detailed in Table 1. Then, we construct bipartite graphs $\{\mathbf{G}^{(v)}\}_{v=1}^{V}$ by solving Eq. (3) with neighborhood size $k = 7$ fixed across views. ESTIM incorporates three regularization parameters: λ, β_λ, and ρ. In our experimental setup, λ is fixed at V^2, while ρ and β_λ are tuned over the ranges $\{10^{-1}, 1, \cdots, 10^6\}$ and $\{2^0, ..., 2^6\}$, respectively. We employ Accuracy (ACC), Precision (Prec), and F1-score (F1) as evaluation metrics. Each experiment was independently repeated ten times and their mean values with standard deviation are reported.

4.2 Main Results

To evaluate the effectiveness of ESTIM, we performed a comparative analysis against baseline methods in two critical scenarios: missing views and scarce labels. For the view missing issue, we follow [49] to randomly select View Missing Ratio (VMR%) examples in dataset as incomplete examples, which randomly miss $1\sim V-1$ views. On this basis, we randomly select Label Annotation Ratio (LAR%) examples of each class as labeled data to simulate the label scarcity setting.

[1] https://www.vision.caltech.edu/datasets/cub_200_2011/
[2] https://archive.ics.uci.edu/dataset/72/multiple+features
[3] https://archive.ics.uci.edu/ml/datasets/Multiple+Features
[4] http://yann.lecun.com/exdb/mnist/
[5] http://www-cvr.ai.uiuc.edu/ponce_grp/data/scene_categories/scene_categories.zip
[6] https://data.caltech.edu/records/20086
[7] https://cvml.ista.ac.at/AwA/
[8] https://www.inaturalist.org/observations/export

Table 2: ACC results of the compared methods on eight datasets under different view-missing ratios (VMRs) when the label annotation ratio (LAR) is fixed at 5%.

Methods	10%	30%	50%	70%	90%	10%	30%	50%	70%	90%
	CUB					**UCI-Digit**				
FMSSL	66.03±8.27	56.82±3.73	51.46±3.63	44.71±5.78	36.07±5.23	85.38±3.53	80.32±3.07	72.45±3.09	54.78±3.17	34.36±7.72
AMGL	67.65±4.07	63.59±5.75	60.68±3.92	56.87±2.96	54.43±5.64	90.82±1.16	90.23±1.31	89.92±1.46	89.26±0.99	85.15±1.34
AMSC	67.05±1.96	61.22±1.94	36.70±3.29	36.70±1.43	23.78±1.22	94.55±0.30	81.43±1.15	59.13±0.43	36.42±1.06	27.69±0.43
LPvote	57.08±3.28	55.10±4.35	53.35±3.27	55.07±4.08	47.17±4.51	92.00±0.87	87.73±0.54	85.39±1.14	82.34±1.79	75.28±2.34
MLAN	67.52±2.53	55.11±5.95	53.17±5.89	49.42±5.38	46.49±5.15	85.29±1.84	80.36±2.45	79.93±4.01	78.71±4.14	80.01±2.98
SLIM	74.67±2.95	70.21±4.45	65.49±3.78	62.00±4.73	48.44±2.99	90.56±0.66	86.49±1.37	82.96±0.99	74.47±1.39	62.78±2.23
SMIR	56.52±10.15	42.03±6.40	32.65±3.00	28.68±3.58	26.26± 3.77	69.76±11.08	54.80±7.76	41.21±4.03	32.64±3.57	28.96±3.37
MVAR	63.54±3.31	43.98±3.65	29.65±5.73	21.19±3.82	19.16±3.16	58.10±3.65	62.37±7.36	57.58±6.16	53.62±5.73	46.32±3.36
FMSEL	61.56±5.49	47.86±4.55	44.00±3.51	38.04±4.52	42.30±4.74	92.97±1.39	85.81±1.48	76.02±2.76	66.70±1.99	60.96±2.68
CFMSC	64.00±0.00	57.05±0.00	60.24±0.00	54.95±0.00	54.84±0.00	93.28±0.00	91.89±0.00	90.55±0.00	88.05±0.00	79.97±0.00
GEGCN	67.10±1.33	65.01±1.85	64.45±1.27	62.20±0.34	60.66±1.26	94.73±1.89	90.87±1.72	87.89±1.27	85.78±1.26	78.35±1.85
ESTIM	**81.77±2.24**	**82.26±2.83**	**83.42±3.69**	**83.35±3.21**	**77.01±5.22**	**95.55±0.71**	**96.45±0.61**	**97.92±0.38**	**98.02±0.48**	**96.45±1.32**
Methods	**Handwritten**					**MINST4**				
FMSSL	73.58±7.53	63.62±14.73	50.31±3.93	38.80±8.35	19.20±2.48	90.12±0.32	89.14±0.55	86.96±0.55	84.75±1.10	75.38±0.59
AMGL	89.05±1.19	87.15±2.06	89.05±2.49	80.00±2.02	75.78±2.87	91.97±0.28	91.66±0.26	91.18±0.20	90.57±0.34	89.70±0.46
AMSC	95.61±0.18	88.47±0.27	66.96±1.02	32.87±0.69	12.93±0.12	91.41±0.12	84.66±0.09	69.03±0.38	53.37±0.61	44.00±0.75
LPvote	93.43±0.90	91.34±1.08	86.11±1.16	80.06±0.84	69.71±1.13	91.39±0.18	91.28±0.33	91.10±0.20	90.60±0.30	90.62±0.30
MLAN	76.17±1.57	59.48±1.42	48.72±3.51	49.18±2.12	55.61±2.66	90.12±0.57	89.73±1.18	89.44±0.34	89.68±0.83	89.80±0.61
SLIM	90.93±1.50	89.96±1.65	83.27±0.94	71.50±2.54	48.44±2.10	89.01±0.43	88.65±0.39	88.41±0.45	87.82±0.47	85.43±0.58
SMIR	65.72±5.75	51.12±2.18	29.95±3.03	26.39±0.60	22.45±0.10	84.14±0.13	70.88±0.11	58.80±1.89	51.52±0.21	46.49±0.32
MVAR	74.28±6.95	70.88±12.64	69.24±14.40	61.04±6.34	42.44±2.56	88.02±0.39	87.42±0.60	85.59±0.74	83.39±0.17	81.19±2.12
FMSEL	91.34±1.11	84.84±1.16	65.65±1.55	45.80±1.43	32.80±1.10	90.62±0.15	90.26±1.07	88.16±0.98	86.96±1.65	86.94±1.34
CFMSC	89.91 ±0.00	89.33±0.00	86.70±0.00	83.24±0.00	77.38±0.00	91.87±0.00	91.85±0.00	91.11±0.00	90.54±0.00	89.29±0.00
GEGCN	93.08±0.36	92.70±0.13	84.40±1.25	82.98±0.42	87.40±1.93	88.42±0.36	89.72±0.16	89.62±0.55	88.96±0.22	89.57±0.59
ESTIM	**96.02±0.59**	**98.03±0.32**	**98.52±0.41**	**98.53±0.39**	**96.46±1.09**	**92.65±0.32**	**92.35±0.25**	**93.28±0.14**	**93.78±0.16**	**93.92±0.43**
Methods	**Scene15**					**Caltech101**				
FMSSL	24.88±2.68	23.16±1.85	22.13±2.28	19.00±1.84	14.03±1.94	27.55±0.65	24.38±0.53	22.86±1.57	16.06±0.94	20.82±1.03
AMGL	42.48±1.70	40.66±1.52	38.09±1.70	36.94±0.99	33.13±1.50	30.25±0.30	23.71±0.44	19.74±0.96	16.85±0.38	12.98±0.76
AMSC	47.15±0.67	31.76±0.35	24.23 ±0.56	14.70±0.70	10.60±0.17	32.38±0.24	29.17±0.19	20.77±0.25	11.63±0.22	6.74±0.24
LPvote	44.67±1.31	42.83±0.87	42.75±0.66	40.86±1.06	36.62±1.25	28.38±0.39	28.28±0.29	27.79±0.56	26.89±0.67	25.11±0.79
MLAN	17.28±4.92	16.53±5.35	16.26±4.34	16.86±4.69	17.04±6.97	27.87±0.49	24.15±0.57	20.91±0.92	20.72±0.83	22.73±0.98
SLIM	21.49±7.55	22.08±7.76	21.34±7.49	18.42±6.47	14.57±5.12	14.75±5.18	14.63±5.16	14.85±5.22	12.71±4.46	12.50±4.76
SMIR	35.08±0.55	25.84±0.83	16.80±1.77	16.14±0.68	15.89±1.27	14.46±1.41	9.28±0.62	7.60±0.76	7.24±0.72	5.10±0.25
MVAR	53.36±1.98	45.15±1.61	39.16±1.58	31.93±0.76	27.92±2.19	38.45±0.42	32.94±1.84	24.61±1.35	23.46±2.35	20.11±2.01
FMSEL	55.17±1.08	50.18±1.52	45.64±1.05	38.47±1.38	31.85±1.08	38.45±0.42	32.94±1.84	24.61±1.35	23.46±2.35	20.11±2.01
CFMSC	45.66±0.00	42.15±0.00	39.27±0.00	34.22±0.00	31.36±0.00	26.73±0.00	27.63±0.00	24.06±0.00	17.65±0.00	13.98±0.00
GEGCN	48.56±2.49	46.53±0.62	42.14±1.07	38.04±1.52	42.30±4.74	21.92±0.70	22.10±0.71	22.07±0.43	22.11±0.33	22.34±0.21
ESTIM	**55.20±1.31**	**54.26±1.13**	**55.74±1.40**	**54.51±1.35**	**47.88±1.41**	**43.92±1.19**	**40.49±0.50**	**37.16±0.68**	**32.40±0.75**	**27.09±1.30**
Methods	**AwA**					**Animal**				
FMSSL	27.53±0.16	17.11±2.18	16.63±2.84	14.58±2.24	8.38±1.67	12.04±1.17	11.45±0.79	12.01±0.32	7.94±2.99	9.34±0.54
AMGL	65.62±0.47	62.11±1.38	59.55±1.10	55.97±0.51	50.81±1.15	16.61±0.77	16.35±0.13	14.41±0.18	14.82±0.37	12.18±0.51
AMSC	65.87±0.74	48.96±0.75	33.88±1.04	27.08±0.42	19.69±0.60	18.43±0.28	15.59±0.36	12.95±0.18	7.90±0.20	5.76±0.10
LPvote	50.89±1.71	52.52±1.23	53.08±0.77	53.16±0.82	47.60±1.29	16.84±0.34	17.71±0.42	17.99±0.44	16.87±0.44	15.59±0.59
MLAN	40.12±3.68	24.59±1.48	25.16±3.04	24.30±2.86	13.99±2.30	15.02±0.73	14.56±0.83	12.90±0.96	12.96±0.72	11.55±1.16
SLIM	70.68±0.85	64.70±0.83	61.44±0.75	58.88±0.23	52.55±0.91	16.85±5.92	18.71±6.57	17.38±6.11	16.92±5.94	15.72±5.52
SMIR	32.36±3.05	27.46±1.63	29.42±2.37	22.86±2.71	23.62±0.97	13.78±2.13	12.87±2.67	11.60±1.82	10.73±0.69	10.21±0.49
MVAR	73.93±0.89	66.19±0.57	60.29±0.22	56.15±0.61	53.45±1.53	22.65±0.80	21.01±1.18	16.45±0.81	13.25±0.26	10.72±0.35
FMSEL	72.29±8.20	61.26±1.05	53.68±4.17	50.50±1.17	48.21±1.93	23.94±1.03	20.31±0.84	17.11±0.74	14.87±0.50	14.59±0.38
CFMSC	65.31±0.00	64.83±0.00	61.21±0.00	56.79±0.00	52.25±0.00	14.15±0.00	17.55±0.00	16.56±0.00	16.46±0.00	14.10±0.00
GEGCN	69.64±1.49	66.37±1.07	65.38±1.59	62.98±2.26	69.35±2.52	23.46±0.43	23.41±0.46	22.65±0.48	19.64±0.81	15.99±0.62
ESTIM	**74.54±1.16**	**72.89±1.46**	**76.18±0.69**	**76.95±0.68**	**73.75±0.91**	**25.84±0.43**	**25.97±0.44**	**25.29±0.34**	**20.81±3.27**	**16.59±1.78**
win/tie/lose	40/0/0	40/0/0	40/0/0	40/0/0	40/0/0	40/0/0	40/0/0	40/0/0	40/0/0	40/0/0

Specifically, from view missing perspective, Table 2 present the compares performance of different methods across eight datasets under varying VMRs with range from (10% ∼ 90%) when fixed LAR as 5%. From the results, we can observe that: (1) Compared to AMSC designed for multi-view semi-supervised learning, GMvSSL methods such as MLAN and AMGL with DMF achieve better performance on CUB and UCI-Digit. These results further verifies that AMSC overlooks that missing samples can distort local structures, thereby misguiding label propagation, and thus reducing performance. (2) ESTIM demonstrates significant performance advantages across all eight datasets when compared to contemporary multi-view learning frameworks. It substantially outperforms graph-based methods, in particular, on the CUB dataset, ESTIM consistently outperforms the second-best method by 15% in classification ACC across all VMRs. (3) Unlike IMvSSL baselines SLIM that ignore missing samples, the tensorial imputation strategy makes ESTIM more robust. For instance, ESTIM exhibits minimal degradation of merely 4. 76% in CUB from 10% to 90% VMR. In contrast,

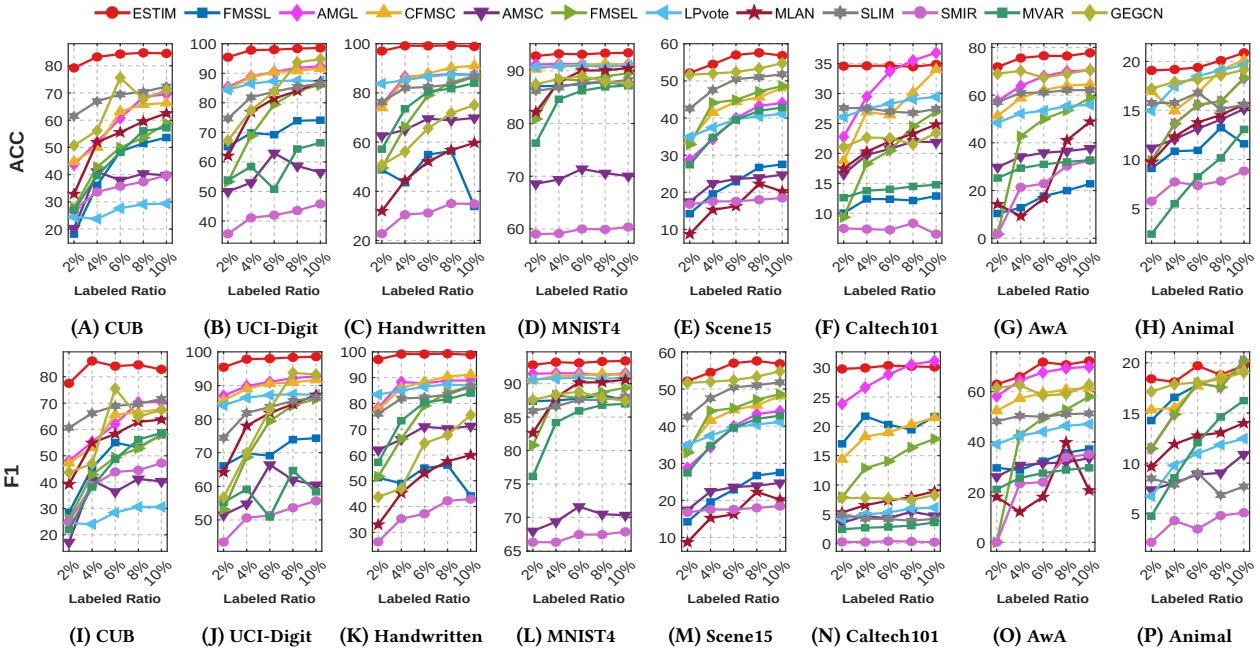


Figure 2: ACC and F1 results on eight datasets with LAR varying in {2%, 4%, 6%, 8%, 10% } when VMR is 50%.

FMSSL's performance in CUB worsened by 29.96%. Compared to early fusion-based GMvSSL methods such as MLAN, FMSEL and CFMSC, ESTIM shows consistent superiority in all cases. This is because ESTIM builds its fusion mechanism with a consensus graph, which is resistant to the structural distortions caused by the missing samples of multi-view data.

From the scarce labels perspective, Figure 2 shows the ACC and F1 results under different LARs (2%, 4%, 6%, 8%, 10%) with VMR fixed as 50%. The remaining results are present in Appendix. From the results, we can observe that: Compared to early fusion-based GMvSSL methods like MLAN, FMSEL and CFMSC, ESTIM achieves consistent superiority across almost all scenarios. This indicates that the consensus graph fusion in ESTIM can effectively mitigate the adverse effects of structural distortions induced by missing samples on label propagation. Besides, it is obvious that all of methods perform better with the increase of LAR in almost all cases. Furthermore, the performance of ESTIM when fixed LAR to 2% is comparable to that of most baselines under higher LARs, demonstrating its higher label utilization efficiency.

4.3 Ablation Studies

To evaluate the contribution of ESTIM's three core components: consensus graph fusion, weight coefficient and tensorial imputation, we perform ablation studies under a 50% VMR with LAR fixed at 5%. In the consensus graph fusion module, we denote the view fusion module as "w/o $\mathbf{G}_v$", where × indicates the exclusion of the permutation matrix for anchor alignment. In addition, For the weight coefficient component, we denote the adaptive update mechanism as "w/o α_v", where × signifies fixing all view weights to equal values by freezing α_v updates. For the tensorial imputation component, we update process of G_{ψ_v} from the optimization algorithm, marked as "w/o TI", where × indicates the removal of G_{ψ_v} updates. Table 3 demonstrate that: (1) The result of experiment (A), (B), (C) indicates that the contribution of each component to the performance demonstrates the efficacy of the ESTIM approach. (2) According to the result of experiments (D), (E), (F), we observed that the classification performance decreased even more when two components of ESTIM are not utilized. The consensus graph fusion module uses complementary structural information among multiple views and the weight coefficient is designed to stress the contribution of diverse views. Both of these directly determine the quality of the fused graph **P**. Therefore, without these two parts, the classification drops the most. (3) The results of experiment (G) show that when all components are installed, our proposed model obtain the optimal performance, which verify that each part of ESTIM plays a crucial role in performance improvement.

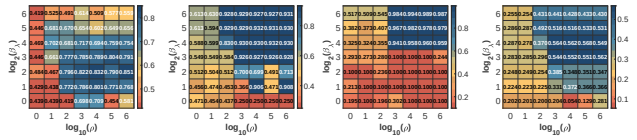
Table 3: Ablation Study on Four Datasets

	$\mathbf{G}_v$	α_v	TI	ACC			
				CUB	UCI-Digit	MNIST4	Scene15
(A)	√	√	×	55.08	93.78	83.71	45.35
(B)	√	×	√	77.19	97.15	93.24	45.49
(C)	×	√	√	44.38	54.26	59.34	43.62
(D)	×	×	√	43.33	52.26	59.74	42.48
(E)	×	√	×	44.74	52.10	61.13	43.01
(F)	√	×	×	49.12	94.73	86.29	45.71
(G)	√	√	√	80.87	97.78	93.28	55.74
	$\mathbf{G}_v$	α_v	TI	F1			
				CUB	UCI-Digit	MNIST4	Scene15
(A)	√	√	×	49.25	93.68	83.63	18.97
(B)	√	×	√	76.46	97.15	93.19	17.20
(C)	×	√	√	43.05	53.96	58.69	16.40
(D)	×	×	√	42.14	51.92	59.46	14.54
(E)	×	√	×	43.89	51.74	60.63	15.87
(F)	√	×	×	48.29	94.73	86.32	16.91
(G)	√	√	√	78.84	97.78	93.25	49.78

4.4 Model Analysis

In this subsection, we perform model analysis under VMR=50% and LAR=5%, focusing on convergence characteristics, parameter sensitivity to β_λ and ρ, and scalability to large scale data.

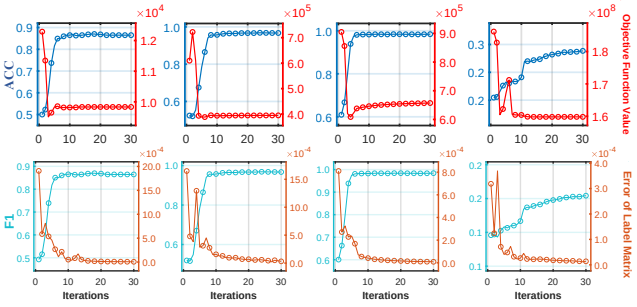
Parameter sensitivity. There are two tunable parameters: the regularization parameter β_λ to regulate the connectivity of the fused graph and ρ for the tensorial imputation. In this section, experiments are conducted on four datasets to evaluate the impact of these two parameters on classification performance by tuning β_λ and ρ in the sets $\{2^0, 2^1, \ldots, 2^6\}$ and $\{10^0, \ldots, 10^6\}$, respectively. ACC results are summarized in Figure 3. Compared to ρ, β_λ exerts more pronounced impacts on ESTIM, indicating that appropriate connectivity of the fused graph can further enhance classification performance. In particular, ESTIM demonstrates stability with high performance within $\log_2(\beta_\lambda) \in [4, 6]$ and $\log_{10}(\rho) \in [2, 5]$ in diverse datasets, maintaining generalization ability.



(A) CUB (B) UCI-Digit (C) Handwritten (D) Scene15

Figure 3: Parameter sensitivity of ESTIM

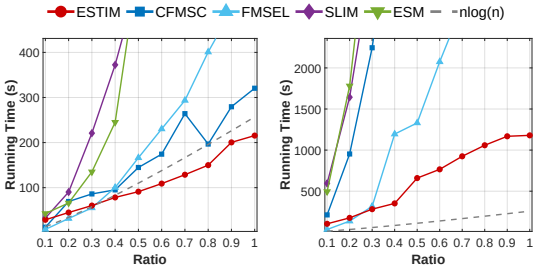
Convergence Study. To reveal the convergence behavior of our ESTIM, we calculate the objective function value, the error value of the soft label matrix $\mathbf{F}$ and its classification performance during iteration in Fig. 4. As observed, the error curves consistently exhibit early oscillations, subsequently undergoing a rapid decrease until convergence. Meanwhile, the objective function value decrease rapidly on most converging to a stable value, which reflects excellent adaptability and generalization performance. In addition, ESTIM's classification performance substantially improves during its iterations, indicating its convergence of proposed algorithm.



(A) CUB (B) UCI-Digit (C) Hnadwritten (D) Animal

Figure 4: The iterative error and classification performance of ESTIM during optimization.

Scalability Study. To validate the scalability of our ESTIM to large-scale datasets in Subsection 3.3, we conduct experiments with different numbers of samples. Meanwhile, to control for the biases of different data sets, we systematically sampled the Animal (11673 samples) and MNISTMV (70000 samples) datasets in proportions of {10%, 20%, ..., 100%}. In each proportion, we run methods independently ten times under VMR = 50% when LAR is fixed at 5% and report their average computational time in Figure 5. From the results, we observe that the growth trend of ESTIM's running time with increasing data scale closely matches our theoretical analysis. Meanwhile, compared to other GMvSSL methods: SLIM, CFMSC and FMSEL, ESTIM needs much lower running times than all competing methods when the number of instances on training exceeding 5,000 samples, providing experimental validation for its applicability to large-scale datasets.



(A) Animal (B) MINSTMV

Figure 5: Scalability versus the large-scale of samples

5 Conclusion

In this paper, we propose a unified framework ESTIM to address the critical challenge of dual missing problems of multi-view features and labels. Specifically, the correspondences of each view are characterized through bipartite graph inside each view for recovery, instead of sample-level similarity matrices. Meanwhile, we design a tensorial imputation method to recover missing sample similarities, leveraging high-order consistent correlations across views with tensor nuclear norm regularization. On the basis of tensorial imputation completion, we construct a universal consensus graph fusion framework to learn a robust consensus graph for label propagation. Finally, experimental results on several public datasets validate the superiority and effectiveness of ESTIM. In the future, we will explore a collaborative learning mechanism between anchor selection and optimal graph construction to enhance the quality of labels.

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A Notations

Assume that $\mathbf{X}=[\mathbf{x}_1,\ldots,\mathbf{x}_n]^\top\in\mathbb{R}^{n\times d}$ is a single view data matrix of d dimensional with n samples, where the first $\ell(\ll n)$ samples are labeled in c classes as $\{y_i\}_{i=1}^{\ell}$. Considering the incomplete multi-view data, let $\mathbf{O}^{(v)}\in\{0,1\}^{n\times n}$ be the diagonal indicator matrix of the existing/missing state in the v-th view, where $o_{ii}^{(v)}=0$ denotes the i-th instance missing its v-th view features and $o_{ii}^{(v)}=1$ otherwise. The main notations of this paper are summarized in Table 4.

B Theoretical Analysis

To prove the convergence of Algorithm 1, we first introduce an important lemma.

Table 4: Summary of Notations

Notations	Annotation
$x, \mathbf{x}, \mathbf{X}, \mathcal{X}$	scalar, vector, matrix, and tensor
$X_{ij}, \mathbf{X}(i,j)$	the element in the i-th row and j-th column of $\mathbf{X}$
$\mathbf{X}_{i\cdot}$	the i-th row vector of $\mathbf{X}$
$\mathcal{X}^k$	the k-th frontal slice of tensor $\mathcal{X}$
$\mathcal{X}_f$=fft($\mathcal{X}$, [], 3)	the fast Fourier transformation (FFT)
n, V, m, ℓ, c	the number of samples, views, anchors, labeled samples, and classes
$\mathbf{1}_n$=$[1,\ldots,1]^\top$	the all-ones column vector with n elements
$\mathbf{I}_n \in \mathbb{R}^{n\times n}$	the n dimensional identity matrix
$\mathbf{X}_v \in \mathbb{R}^{n\times d_v}$	the d_v dimensional feature matrix of v-th view
$\mathbf{x}_i^v \in \mathbb{R}^{d_v\times 1}$	the i-th sample of v-th view
$\{y_i \in [c]\}_{i=1}^{\ell}$	the label set of ℓ labeled samples
$\mathbf{Y} \in \{0,1\}^{n\times c}$	the one-hot label matrix of n samples
$\mathbf{W}$ ($\mathbf{W}_v$)$\in[0,1]^{n\times n}$	the symmetric similarity matrix among n samples (of v-th view)
$\mathbf{D}_W$	the diagonal degree matrix of $\mathbf{W}$ with entries $d_i = \sum_t \mathbf{W}_{it}$
$\mathbf{L}_W$=$\mathbf{D}_W$−$\mathbf{W}$	the Laplacian matrix of $\mathbf{W}$
$\{\mathbf{a}_j^v \in \mathbb{R}^{d_v\times 1}\}_{j=1}^m$	the set of m anchors in v-th view
$\mathbf{G}_v \in \Delta_m^n$	the anchor-based bipartite matrix of v-th view
$\mathbf{O}^{(v)}$	the index sets of existing and missing samples in v-th view
$\mathbf{P} \in \Delta_m^n$	the fused/consensus anchor-based bipartite graph
$\mathbf{T}_v \in \mathbb{R}^{m\times m}$	the permutation matrix of v-th view
$\alpha^\top \in \Delta_V^1, \alpha_v \in [0,1]$	the weight coefficient vector and the weight coefficient of v-th view
$\mathbf{W}_P \in \mathbb{R}^{(n+m)\times(n+m)}$	the similarity matrix among fused n samples and m anchors
$\hat{\mathbf{F}}$=$[\mathbf{F};\mathbf{Q}] \in \mathbb{R}^{(n+m)\times c}$	the soft label matrix of fused samples and anchors, where $\mathbf{F}$ is the soft label matrix of samples and $\mathbf{Q}$ is for anchors
$\hat{\mathbf{Y}}$=$[\mathbf{Y};\mathbf{0}]$	the one-hot true label matrix of samples and anchors
$\mathbf{B}, \hat{\mathbf{B}}$	the diagonal regularization matrix
$\Phi(\cdot)$	the merging and rotating operator
$\|\cdot\|_F, \|\cdot\|_2$	the Frobenius norm and ℓ_2 norm
$\|\cdot\|_\circledast$	the t-SVD based tensor nuclear norm
Δ_m^n	$\{\zeta \in \mathbb{R}^{n\times m} \mid \zeta \mathbf{1}_m = \mathbf{1}_n, \zeta \geq 0\}$

Lemma B.1. [18] *Suppose that $F : \mathbb{R}^{m\times n} \mapsto \mathbb{R}$ is defined as $F(X) = f\circ\sigma(X) = f(\sigma_1(X), ..., \sigma_r(X))$,where $X = UDiag(\sigma(X))V^\top$ is the normal SVD of matrix $X \in \mathbb{R}^{mn}, r = min(m,n)$, and $f(\cdot) : \mathbb{R}^r \mapsto \mathbb{R}$ be differentiable and absolutely symmetric at $\sigma(X)$ Then the subdifferential of $F(X)$ at x is:*

$$\frac{\partial F(X)}{\partial X} = UDiag(\partial f(\sigma(X)))V^\top, \tag{27}$$

where $\partial f(\sigma(X)) = (\frac{\partial f(\sigma_1(X))}{\partial X}, ..., \frac{\partial f(\sigma_r(X))}{\partial X})$.

Proof: In order to prove the boundedness property of the sequence $\{\{\mathbf{G}_{\psi_v}^{(k)}, \mathbf{T}_v^{(k)}, \alpha_v^{(k)}\}_v^V, \hat{\mathbf{F}}^{(k)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k)}, \mathcal{U}^{(k)}\}_{k=1}^\infty$, we formulate the following Lyapunov function:

$$\begin{aligned}
&\mathcal{V}_{\eta_k}(\{\mathbf{G}_{\psi_v}^{(k)}, \mathbf{T}_v^{(k)}\ \alpha_v^{(k)}\}_v^V, \hat{\mathbf{F}}^{(k)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k)}, \mathcal{U}^{(k)}\}) \\
&= \mathcal{J}(\{\mathbf{G}_{\psi_v}^{(k)}, \mathbf{T}_v^{(k)}\ \alpha_v^{(k)}\}_v^V, \hat{\mathbf{F}}^{(k)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k)}, \mathcal{U}^{(k)}\}) + \|\hat{\mathbf{Y}}\|_F^2. \\
&= \mathrm{Tr}\left((\mathbf{I}_{n+m} + \hat{\mathbf{D}})\hat{\mathbf{F}}^{(k)\top}\hat{\mathbf{F}}^{(k)}\right) - \mathrm{Tr}\left(\hat{\mathbf{F}}^{(k)\top}\hat{\mathbf{W}}_P{}^{(k)}\hat{\mathbf{F}}^{(k)}\right) - 2\mathrm{Tr}\left(\hat{\mathbf{D}}\hat{\mathbf{Y}}\hat{\mathbf{F}}^{(k)\top}\right) \\
&- \lambda\sum_v \alpha_v^{(k)}\mathrm{Tr}\left(\mathbf{P}^{(k)\top}(\mathbf{G}_v^{(k)}\mathbf{T}_v^{(k)})\right) + \beta_\lambda\|\mathbf{P}^{(k)}\|_F^2 + \rho\|\mathcal{H}^{(k)}\|_\circledast \\
&+ \langle\mathcal{U}^{(k)}, \mathcal{G}^{(k)} - \mathcal{H}^{(k)}\rangle + \frac{\eta}{2}\|\mathcal{G}^{(k)} - \mathcal{H}^{(k)}\|_F^2,
\end{aligned} \tag{28}$$

where $\|\hat{\mathbf{Y}}\|_F^2$ is a constant. To prove the multiplier sequence $\{\mathcal{U}^{(k)}\}_{k=1}^\infty$ is bounded. We derive the first-order optimality condition of $\mathcal{G}$ by the updating rule:

$$\partial\|\mathcal{H}^{(k+1)}\|_\circledast = \mathcal{U}^{(k)} + \eta_k(\mathcal{G}^{(k+1)} - \mathcal{H}^{(k+1)}) = \mathcal{U}^{(K+1)}, \tag{29}$$

where $\mathcal{U}*\mathcal{S}*\mathcal{V}^\top$ be the t-SVD of tensor $\mathcal{H}$. According to Lemma B.1, we have:

$$\begin{aligned}
\left\|\partial\|\mathcal{H}^{(k+1)}\|_\circledast\right\|_F^2 &= \left\|\frac{1}{n}\mathcal{U} * \mathrm{ifft}(\partial\mathcal{S}_f, [], 3) * \mathcal{V}^\top\right\|_F^2 \\
&= \frac{1}{n^3}\|\partial\mathcal{S}_f\|_F^2 \leq \frac{1}{n^3}\sum_{i=1}^{n}\sum_{j=1}^{\min(m,V)} 1,
\end{aligned} \tag{30}$$

which implies $\partial\|\mathcal{H}^{(k+1)}\|_\circledast$ is bounded. From Eq. (29), we can infer that the sequence $\{\mathcal{U}^{(k)}\}_{k=1}^\infty$ is also bounded.

Note that all subproblems of the ADMM procedure, *i.e.*, Eq. (11), Eq. (12), Eq. (24), Eq. (16), Eq. (19), and Eq. (20), are closed, proper and convex, we can infer that:

$$\begin{aligned}
&\mathcal{V}_{\eta_k}(\{\mathbf{G}_{\psi_v}^{(k+1)}, \mathbf{T}_v^{(k+1)}, \alpha_v^{(k+1)}\}_v^V, \hat{\mathbf{F}}^{(k+1)}, \mathbf{P}^{(k+1)}, \mathcal{H}^{(k+1)}, \mathcal{U}^{(k+1)}) \\
&\leq \mathcal{V}_{\eta_k}(\{\mathbf{G}_{\psi_v}^{(k)}, \mathbf{T}_v^{(k)}\alpha_v^{(k)}\}_v^V, \hat{\mathbf{F}}^{(k)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k)}, \mathcal{U}^{(k)}) \\
&= \mathcal{V}_{\eta_{k-1}}(\{\mathbf{G}_{\psi_v}^{(k)}, \mathbf{T}_v^{(k)}\alpha_v^{k}\}_v^V, \hat{\mathbf{F}}^{(k)}, \mathcal{H}^{(k)}, \mathcal{U}^{(k+1)}, \mathbf{P}^{(k)}) \\
&+ \frac{\eta_k + \eta_{k-1}}{2\eta_{k-1}^2}\|\mathcal{U}^{(k)} - \mathcal{U}^{(k+1)}\|_F^2.
\end{aligned} \tag{31}$$

By summing the right-hand side of Eq. (31) from k=1 to n, we have:

$$\begin{aligned}
&\mathcal{V}_{\eta_k}(\{\mathbf{G}_{\psi_v}^{(k+1)}, \mathbf{T}_v^{(k+1)}, \alpha_v^{(k+1)}\}_v^V, \hat{\mathbf{F}}^{(k+1)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k+1)}, \mathcal{U}^{(k)}) \\
&\leq \mathcal{V}_{\eta_0}(\{\mathbf{G}_{\psi_v}^{(1)}, \mathbf{T}_v^{(1)}, \alpha_v^{(1)}\}_v^V, \hat{\mathbf{F}}^{(1)}, \mathbf{P}^{(1)}, \mathcal{H}^{(1)}, \mathcal{U}^{(0)}) \\
&+ \sum_{k=1}^{n}\frac{\eta_k + \eta_{k-1}}{2\eta_{k-1}^2}\|\mathcal{U}^{(k)} - \mathcal{U}^{(k-1)}\|_F^2.
\end{aligned} \tag{32}$$

Since $\sum_{k=1}^n \frac{\eta_k+\eta_{k-1}}{2\eta_{k-1}^2} < \infty$, $\mathcal{V}_{\eta_0}(\{\mathbf{G}_{\psi_v}^{(1)}, \mathbf{T}_v^{(1)}, \alpha_v^{(1)}\}_v^V, \hat{\mathbf{F}}^{(1)}, \mathbf{P}^{(1)}, \mathcal{H}^{(1)}, \mathcal{U}^{(0)})$ is finite, and the sequence $\{\mathcal{U}^{(k)}\}_{k=1}^\infty$ is bounded, we can derive $\mathcal{V}_{\eta_k}(\{\mathbf{G}_{\psi_v}^{(k+1)}, \mathbf{T}_v^{(k+1)}, \alpha_v^{(k+1)}\}_v^V, \hat{\mathbf{F}}^{(k+1)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k+1)}, \mathcal{U}^{(k)})$ is bounded. Eq. (6) suggest that $\mathrm{Tr}\left((\mathbf{I}_{n+m}+\hat{\mathbf{D}})\hat{\mathbf{F}}^{(k+1)\top}\hat{\mathbf{F}}^{(k+1)}\right) - \mathrm{Tr}\left(\hat{\mathbf{F}}^{(k+1)\top}\hat{\mathbf{W}}_{P^{(k+1)}}\ \hat{\mathbf{F}}^{(k+1)}\right) - 2\mathrm{Tr}\left(\hat{\mathbf{D}}\hat{\mathbf{Y}}\hat{\mathbf{F}}^{(k+1)\top}\right) \geq 0$. Therefore, Recall the specific Lyapunov function in Eq. (28), except for $-\alpha_v\mathrm{Tr}\left(\mathbf{P}^{(k+1)\top}(\mathbf{G}_v^{(k+1)}\mathbf{T}_v^{(k+1)})\right)$, the other terms are non-negative. Considering the constraints $\mathbf{P}, \mathbf{G}_v \in \Delta_m^n, \alpha^\top \in \Delta_V^1, \mathbf{T}_v^\top\mathbf{T}_v = \mathbf{I}_n (\forall v \in [V])$, we can infer $\alpha_v\mathrm{Tr}\left(\mathbf{P}^{\star\top}(\mathbf{G}_v^{(k+1)}\ \mathbf{T}_v^{(k+1)})\right) < \infty$. Thus, we can deduce that each term of Eq. (28) is bounded.

The boundedness of $\|\mathcal{H}^{(k+1)}\|_\circledast$ suggests that all singular values of $\mathcal{H}^{(k+1)}$ are bounded. Based on the following equation:

$$\|\mathcal{H}\|_F^2 = \frac{1}{n}\|\mathcal{G}_f^{(k+1)}\|_F^2 = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{\min(m,V)}(\mathcal{S}_f^i(j,j))^2, \tag{33}$$

the boundedness of the sequence $\{\mathcal{H}^{(k)}\}_{k=1}^\infty$ is ensured.

Since $\hat{\mathbf{D}}$ and $\hat{\mathbf{Y}}$ are constants, the boundedness of $\mathrm{Tr}((\hat{\mathbf{F}}^{(k+1)} - \hat{\mathbf{Y}})^\top\hat{\mathbf{D}}(\hat{\mathbf{F}}^{(k+1)} - \hat{\mathbf{Y}}))$ guarantees the boundedness of the sequence $\{\hat{\mathbf{F}}^{(k+1)}\}_{k=1}^\infty$.

Furthermore, according to the update formulas in Eq. (11) and Eq. (20), it could be deduced that these sequences$\{\mathbf{G}_{\psi_v}^{(k)}\}_{k=1}^\infty$ and

$\{\mathbf{T}_v^{(k)}\}_{k=1}^{\infty}$ are bounded. In the end, according to the constraints $\mathbf{P}\in\Delta_m^n$ and $\alpha^\top\alpha = 1$, we can derive that the sequences $\{\mathbf{P}^{(k)}\}_{k=1}^{\infty}$ and $\{\alpha_v^{(k)}\}_{k=1}^{\infty}$ are also bounded.

Assume the sequence $\{\{\mathbf{G}_{\psi_v}^{(k)}, \mathbf{T}_v^{(k)}, \alpha_v^{(k)}\}_v^V, \hat{\mathbf{F}}^{(k)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k)}, \mathcal{U}^{(k)}\}_{k=1}^{\infty}$ is generated by Algorithm 1 during iteration. According to the Weierstrass-Bolzano theorem, there exists at least one accumulation point, denoted as $\{\{\mathbf{G}_{\psi_v}^{(\infty)}, \mathbf{T}_v^{(\infty)}, \alpha_v^{(\infty)}\}_v^V, \hat{\mathbf{F}}^{(\infty)}, \mathbf{P}^{(\infty)}, \mathcal{H}^{(\infty)}, \mathcal{U}^{(\infty)}\}_{k=1}^{\infty}$, such that $\lim_{k\to\infty}\{\{\mathbf{G}_{\psi_v}^{(k)}, \mathbf{T}_v^{(k)}, \alpha_v^{(k)}\}_v^V, \hat{\mathbf{F}}^{(k)}, \mathbf{P}^{(k)}, \mathcal{H}^{(k)}, \mathcal{U}^{(k)}\}_{k=1}^{\infty} = \{\{\mathbf{G}_{\psi_v}^{(\infty)}, \mathbf{T}_v^{(\infty)}, \alpha_v^{(\infty)}\}_v^V, \hat{\mathbf{F}}^{(\infty)}, \mathbf{P}^{(\infty)}, \mathcal{H}^{(\infty)}, \mathcal{U}^{(\infty)}\}_{k=1}^{\infty}$.

From the update formula of $\mathcal{U}$, we have $\mathcal{G}^{(k+1)} - \mathcal{H}^{(k+1)} = (\mathcal{U}^{k+1} - \mathcal{U}^{(k)})/\eta^{(k)}$. Leveraging the boundedness of $\{\mathcal{U}^{(k)}\}_{k=1}^{\infty}$, we obtain $\mathcal{G}^{(\infty)} - \mathcal{H}^{(\infty)} = 0$. Due to the first-order optimality condition of $\mathcal{H}^{(k)}$, we derive $\mathcal{U}^{(\infty)} \in \partial\|\mathcal{H}^{(\infty)}\|_{\circledast}$. Taking into account the closed-form solutions in Eq. (11), Eq. (15), Eq. (18), Eq. (20), Eq. (26), and Theorem 3.1, the accumulation point $(\{\mathbf{G}_{\psi_v}^{(\infty)}, \mathbf{T}_v^{(\infty)}, \alpha_v^{(\infty)}\}_v^V, \hat{\mathbf{F}}^{(\infty)}, \mathbf{P}^{(\infty)}, \mathcal{H}^{(\infty)}, \mathcal{U}^{(\infty)})$ satisfies the corresponding KKT conditions of the problem in Eq. (7).

C Additional Results

Effect of the Number of Anchors. To examine the impact of the anchor number m, we vary its value within the range $\{2^5, 2^6, ..., 2^9\}$ across the four datasets. For each configuration, we report the average running time and classification performance. The complete results are presented in Figure 6, The results show that a clear relationship between the proportion of anchors relative to samples and model performance: accuracy improves as this proportion increases, peaks at an optimal point, and then declines. Accordingly, we set the anchor number m to 5%–10% of the total dataset size that maintains effectiveness across varying data scales.

Effect of the View Number. In our theoretical analysis, we derived the time complexity. To validate this empirically, we conducted experiments on the Animal dataset with four views, evaluating the computational and memory complexity as the number of views increased. The experiments were configured with VMR=50%, LAR=5%, and anchor number of 256. As shown in the table5, the results clearly demonstrate that the empirical trends align with our theoretical analysis: owing to the anchor strategy, the running time and memory usage of ESTIM do not grow excessively with sample size and remain within a manageable range.

Table 5: Time and memory consumption on Animal dataset with varying numbers of views and sample sizes.

Metric	Views	4000	6000	8000	10000
Time (s)	2	26.03	40.49	39.51	66.42
	3	36.41	56.89	50.85	96.96
	4	51.33	59.54	72.53	116.07
Memory (GiB)	2	0.061	0.11	0.15	0.25
	3	0.081	0.14	0.19	0.28
	4	1.02	0.18	0.25	0.33

Empirical Analysis of Time Complexity. To verify the efficiency of ESTIM, we compare its running time with those of six

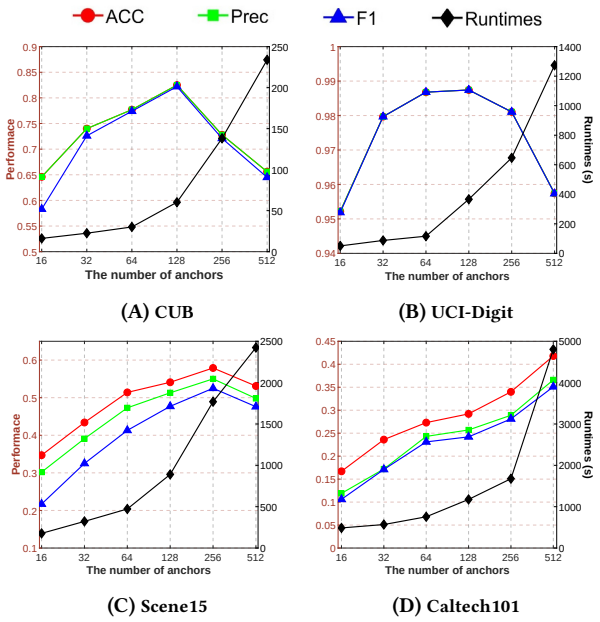


Figure 6: Classification performance of ESTIM with different anchor numbers

state-of-the-art methods across eight datasets under VMR = 50% when LAR is fixed at 5%. The runtime comparisons in Table 6 demonstrate ESTIM's computational superiority through three dimensions: (1) As the data scale increases, the running time of ESTIM shows an approximately linear growth trend. For example, on the CUB dataset , the running time of ESTIM is 2.389 seconds. As the number of samples increases to 11,673 in the Animal dataset, ESTIM's running time rises to 129.7 seconds. The growth aligns with ESTIM's design, which uses anchor-based bipartite graphs to reduce computational complexity. (2) Compared to other tensor-based methods CFMSC, our proposed ESTIM is capable of efficiently processing large-scale datasets. This is attributed to the anchor representation strategy, which reduces the computational complexity of tensor-related operations from $O(n^2)$ to $O(n\log(n))$. (3) Since anchor selection is designed for large-scale scenarios, ESTIM requires more time than adaptive graph learning methods like FMSEL and MLAN on small datasets. Nevertheless, ESTIM significantly outperforms them on large-scale datasets.

Table 6: Running time (seconds) of each method on all datasets with VMR=50% and LAR=5%.

Datasets	FMSSL	MLAN	SLIM	MVAR	CFSMC	FMSEL	ESTIM
CUB	3.825	0.4953	4.296	0.7560	3.025	0.3509	2.389
UCI-Digit	24.77	4.296	59.48	6.382	24.37	3.120	8.735
Handwritten	38.67	9.962	157.0	28.60	44.58	20.79	11.70
MNIST4	94.48	18.86	201.6	18.13	15.43	20.20	30.04
Scene15	115.6	139.1	193.9	25.46	58.17	19.04	25.33
Caltech101	428.8	103.9	1180	136.7	416.3	115.9	51.67
AwA	587.4	497.2	740.9	357.4	628.9	358.5	217.6
Animal	788.6	151.3	1594	237.4	959.5	595.1	129.7